

NIGHT-BEFORE REVIEW

AP Physics C: Mechanics

Unit 1 Kinematics Study Guide

Vectors | Calculus | Motion Graphs | Relative Motion | Projectiles

The central idea

Kinematics describes motion. Always begin by choosing a reference frame and coordinate directions, then connect position, velocity, and acceleration using slopes, areas, derivatives, integrals, or constant-acceleration equations.

What Unit 1 covers

Topic	You should be able to...
1.1 Scalars and vectors	Resolve, combine, and interpret vector quantities.
1.2 x , v , and a	Use definitions, derivatives, integrals, and initial conditions.
1.3 Representing motion	Translate among words, diagrams, graphs, and equations.
1.4 Relative motion	Convert measurements between inertial reference frames.
1.5 Two-dimensional motion	Analyze components independently, especially projectiles.

Prepared for the September 3, 2026 unit test

Aligned to the 2026 College Board AP Physics C: Mechanics Course and Exam Description. Numerical examples use $g = 10 \text{ m/s}^2$ unless stated otherwise.

Tonight's highest-value plan

Time	Task	Proof that you are done
20 min	Formula and graph recall	Write the formulas and slope/area rules without notes.
25 min	Calculus kinematics	Differentiate $x(t)$; integrate $a(t)$; use initial conditions.
30 min	Projectile motion	Complete one angled and one horizontal launch problem.
25 min	Relative motion	Redo two recorded problems using object minus observer.
25 min	Practice check	Finish the ten questions and correct every mistake.
10 min	Trap review	Explain every common trap aloud.

Tomorrow morning

Spend 10-15 minutes recalling formulas, signs, and graph rules. Do not try to learn a brand-new topic immediately before the test.

Final readiness checklist

- Convert vectors between components and magnitude/direction.
- Distinguish distance from displacement and speed from velocity.
- Differentiate $x(t)$ to find $v(t)$ and $a(t)$.
- Integrate $a(t)$ or $v(t)$ and apply initial conditions.
- Use graph slopes and signed areas correctly.
- Use constant-acceleration equations only when acceleration is constant.
- Separate projectile motion into independent x and y components.
- Find relative velocity using object minus observer.
- Report units, signs, and directions.

Formula sheet: know these cold

Definitions and averages

Quantity	Definition	Type
Position r	Location relative to an origin	Vector
Distance	Total path length	Scalar
Displacement Δr	$r_f - r_i$	Vector
Speed	Magnitude of velocity	Scalar
Velocity v	Rate of change of position	Vector
Acceleration a	Rate of change of velocity	Vector

$$v_{\text{avg}} = \Delta r / \Delta t \quad | \quad a_{\text{avg}} = \Delta v / \Delta t$$

$$\text{average speed} = \text{total distance} / \Delta t$$

Calculus relationships

$$v = dr/dt \quad | \quad a = dv/dt = d^2r/dt^2$$

$$\Delta r = \int v(t) dt \quad | \quad \Delta v = \int a(t) dt$$

Constant acceleration only

$$v = v_0 + at$$

$$x = x_0 + v_0 t + (1/2)at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

$$x - x_0 = [(v_0 + v)/2]t$$

Decision rule

If acceleration changes with time, position, or velocity, do not use the constant-acceleration equations. Use derivatives or integrals.

1. Scalars, vectors, and signs

Components

For an angle theta measured from the positive x-axis:

$$A_x = A \cos(\text{theta}) \quad | \quad A_y = A \sin(\text{theta})$$

$$A = \text{sqrt}(A_x^2 + A_y^2) \quad | \quad \text{theta} = \text{atan2}(A_y, A_x)$$

Angle from vertical

If the angle is measured from the vertical, the vertical component is adjacent and uses cosine; the horizontal component is opposite and uses sine.

Signs are physical information

- After choosing east/right/up as positive, west/left/down components are negative.
- A negative velocity means motion in the negative direction.
- A negative acceleration means acceleration points in the negative direction; it does not automatically mean slowing down.
- Use components to add or subtract vectors; do not add magnitudes unless the vectors point in the same direction.

Velocity and acceleration signs

Velocity sign	Acceleration sign	Speed
Same	Same	Increases
Opposite	Opposite	Decreases

Quick example

Walk 5 m east, then 2 m west. Distance = 7 m, but displacement = +3 m east. Average speed uses 7 m; average velocity uses +3 m.

2. Motion graphs and representations

Graph	Slope	Area
Position vs. time	Velocity	No standard kinematics meaning
Velocity vs. time	Acceleration	Displacement
Acceleration vs. time	Rate of change of acceleration	Change in velocity

Position-time graph

- Positive slope means positive velocity; negative slope means negative velocity.
- A horizontal tangent means the object is instantaneously at rest.
- Concave up means positive acceleration; concave down means negative acceleration.
- A steeper slope means a greater speed magnitude.

Velocity-time graph

- Above the axis means positive velocity; below the axis means negative velocity.
- Crossing the axis usually marks a change in direction.
- Slope gives acceleration.
- Signed area gives displacement. Total absolute area gives distance.

Motion diagrams

- Dots represent equal time intervals.
- Increasing spacing means increasing speed; decreasing spacing means decreasing speed.
- Velocity arrows point in the direction of motion.
- Acceleration arrows point in the direction of the change in velocity, which may differ from the direction of motion.

Turning-point trap

At an instantaneous turning point, $v = 0$, but acceleration can be nonzero. At the top of a projectile's path, $v_y = 0$ while $a_y = -g$.

3. Calculus kinematics

Given position $x(t)$: differentiate

Example: $x(t) = 2 + 3t - 4t^2 + t^3$.

$$v(t) = 3 - 8t + 3t^2 \quad | \quad a(t) = -8 + 6t$$

At $t = 2$ s: $v = -1$ m/s and $a = 4$ m/s².

Given acceleration $a(t)$: integrate

Example: $a(t) = 6t$, $v(0) = 2$ m/s, and $x(0) = 0$.

$$v(t) = \text{integral } 6t \, dt = 3t^2 + C$$

The condition $v(0) = 2$ gives $C = 2$, so $v(t) = 3t^2 + 2$.

$$x(t) = \text{integral } (3t^2 + 2) \, dt = t^3 + 2t + C_2$$

The condition $x(0) = 0$ gives $C_2 = 0$.

Acceleration as a function of position

$$a = dv/dt = (dv/dx)(dx/dt) = v(dv/dx)$$

Therefore:

$$\text{integral from } v_0 \text{ to } v \text{ of } v \, dv = \text{integral from } x_0 \text{ to } x \text{ of } a(x) \, dx$$

Integration constants

Every indefinite integration introduces a constant. Use the initial position or velocity to determine it. Missing the constant is one of the most common AP Physics C mistakes.

4. Reference frames and relative motion

Core shortcut: object minus observer

$$v_{\text{object/observer}} = v_{\text{object/ground}} - v_{\text{observer/ground}}$$

Equivalent addition form:

$$v_{A/C} = v_{A/B} + v_{B/C}$$

Why subtraction works

Relative position is the separation between the object's position and the observer's position:

$$r_{\text{object/observer}} = r_{\text{object}} - r_{\text{observer}}$$

Differentiating gives the relative-velocity rule.

One-dimensional check

A car travels east at 10 m/s while the observer travels east at 6 m/s:

$$v_{\text{car/observer}} = 10 - 6 = 4 \text{ m/s east}$$

If the observer travels east at 12 m/s:

$$v_{\text{car/observer}} = 10 - 12 = -2 \text{ m/s} = 2 \text{ m/s west}$$

Two-dimensional method

- Choose x and y directions.
- Resolve both velocities into components.
- Subtract component by component: object minus observer.
- Recombine the result into magnitude and direction.

Inertial frames

Observers whose frames move at constant velocity relative to one another measure the same acceleration for an object. Rotating or accelerating observer frames require additional care.

Recorded examples

Your worked train, airplane, Juan, and Mia problems remain in relative-velocity-solutions.md in the AP Physics folder.

5. Motion in two dimensions and projectiles

Treat perpendicular components independently

$$a_x = 0 \quad | \quad a_y = -g$$

$$x = x_0 + v_{0x}t \quad | \quad v_x = v_{0x}$$

$$y = y_0 + v_{0y}t - (1/2)gt^2 \quad | \quad v_y = v_{0y} - gt$$

For launch angle theta above horizontal:

$$v_{0x} = v_0 \cos(\theta) \quad | \quad v_{0y} = v_0 \sin(\theta)$$

The bridge between axes

Horizontal and vertical motion have separate equations, but the same elapsed time t . Usually solve one axis for time, then use that time in the other axis.

At the highest point

$$v_y = 0, \quad v_x = v_{0x}, \quad a_x = 0, \quad a_y = -g$$

Special case: launch and landing at the same height

$$T = 2v_0 \sin(\theta)/g$$

$$R = v_0^2 \sin(2\theta)/g$$

$$H = v_0^2 \sin^2(\theta)/(2g)$$

Do not use these three special-case formulas unchanged if the launch and landing heights differ.

A reliable test-day process

1. Draw the situation and label the coordinate axes.
2. Choose positive directions and keep them throughout the problem.
3. List known values with signs and units.
4. Identify the requested quantity and its reference frame.
5. Split vectors into components before combining them.
6. Decide whether acceleration is constant.
7. Select derivatives, integrals, graphs, or a constant-acceleration equation.
8. Solve symbolically before inserting numbers when possible.
9. Check units, signs, direction, and physical reasonableness.

Common traps

Trap	Correction
Distance vs. displacement	Average velocity uses displacement; average speed uses total distance.
Negative acceleration	Compare the signs of v and a before deciding whether speed increases or decreases.
Changing acceleration	Do not use constant-acceleration equations when a is not constant.
Integration	Include constants and apply initial conditions.
Angles	Choose sine and cosine based on the axis from which the angle is measured.
Projectiles	Use separate x and y equations connected by the same time.
Projectile peak	Only v_y is zero; a_y remains $-g$.
Graph area	Signed area under $v(t)$ is displacement; absolute area is distance.
Relative velocity	Use object minus observer, in the same reference frame.
Rounding	Keep extra digits until the final result.

Practice check - no notes

1. An object's position is $x(t) = 4 + 5t - 2t^2$. Find its velocity and acceleration at $t = 3$ s.
2. An object has $v(t) = 6 - 2t$ from $t = 0$ to 5 s. Find displacement, distance, average velocity, and average speed.
3. A cart starts from rest and accelerates at 3.0 m/s^2 for 4.0 s. Find final speed and displacement.
4. A car moving at 20 m/s has constant acceleration -5.0 m/s^2 . Find stopping time and distance.
5. A ball is launched from level ground at 20 m/s, 30 degrees above horizontal. With $g = 10 \text{ m/s}^2$, find time of flight, range, and maximum height.
6. A ball rolls horizontally from a 45 m cliff at 10 m/s. With $g = 10 \text{ m/s}^2$, find flight time, range, and impact-speed magnitude.
7. A cyclist moves east at 12 m/s while an observer moves east at 7 m/s. Find the cyclist's relative velocity. Repeat for an observer moving east at 15 m/s.
8. Given $a(t) = 6t$, $v(0) = 2 \text{ m/s}$, and $x(0) = 0$, find $v(2)$ and $x(2)$.
9. At one instant an object has negative velocity and positive acceleration. Is it speeding up or slowing down? Explain.
10. State v_x , v_y , a_x , and a_y at the highest point of an ideal projectile's path.

Before checking answers

For every numerical problem, show your coordinate choice, equation or calculus step, substitution, units, and final direction where applicable.

Practice answers 1-5

1. Position function

Solution

$v(t) = 5 - 4t$ and $a(t) = -4$. At $t = 3$ s: $v = -7$ m/s and $a = -4$ m/s².

2. Displacement versus distance

Solution

The direction changes at $t = 3$ s. $\Delta x = \text{integral from 0 to 5 of } (6 - 2t)dt = 5$ m. The object travels 9 m forward and 4 m backward, so distance = **13 m**. Average velocity = **1.0 m/s**; average speed = **2.6 m/s**.

3. Constant acceleration

Solution

$v = 0 + (3.0)(4.0) = 12$ m/s. $\Delta x = (1/2)(3.0)(4.0)^2 = 24$ m.

4. Braking

Solution

$0 = 20 - 5t$ gives $t = 4.0$ s. Then $0^2 = 20^2 + 2(-5)\Delta x$ gives **$\Delta x = 40$ m**.

5. Angled projectile

Solution

$v_{0x} = 20\cos 30^\circ = 17.3$ m/s and $v_{0y} = 20\sin 30^\circ = 10$ m/s. Therefore **$T = 2.0$ s**, **$R = 34.6$ m**, and **$H = 5.0$ m**.

Practice answers 6-10

6. Horizontal launch

Solution

Vertical motion gives $-45 = -(1/2)(10)t^2$, so $t = 3.0$ s. Range = $(10)(3.0) = 30$ m. At impact $v_x = 10$ m/s and $v_y = -30$ m/s, so speed = $\sqrt{10^2 + 30^2} = 31.6$ m/s.

7. Relative velocity

Solution

Object minus observer. First: $12 - 7 = 5$ m/s east. Second: $12 - 15 = -3$ m/s, so 3 m/s west.

8. Integration

Solution

Integrate and apply the initial conditions: $v(t) = 3t^2 + 2$ and $x(t) = t^3 + 2t$. Thus $v(2) = 14$ m/s and $x(2) = 12$ m.

9. Graph reasoning

Solution

The velocity and acceleration have opposite signs, so the object's speed is **decreasing**. It is slowing down.

10. Projectile peak

Solution

At the highest point: $v_x = v_{0x}$, $v_y = 0$, $a_x = 0$, and $a_y = -g$.

Last reminder

A correct answer needs more than a number. Show the physical model, signs, vector components, units, and direction. If your result contradicts the diagram, revisit the setup before changing the algebra.